

Notes Spring 2025
TTK 4135 Optimization and Control

Adam Godtland Moumen

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1 Week 3

1.1 Assignments

- Exercises, 7/10 approval required
- Helicopter lab
- Matlab assessments

1.2 Grading

- Final exam: 80%
- Matlab assessments: 20%

1.3 Course material

- Numerical Optimization, J.Nocedal and S. J Wright, 2nd ed.
- Errata on BB.
- Note on Merging Optimization and Control, B. Foss and T.A. Heirung.
- Note on Matrix Calculus, T.A. Heirung.

1.4 Motivation

Optimization: Find best solution to a given task (mathematical problem).
For the non general problems numerical techniques must be used. Applied
mathematics built on:

- Convexity and non-convexity
- Global vs. local solution
- Constrained vs. unconstrained problems
- Feasible set

Typical application for Optimization:

- Autonomous vehicles
- Power plants
- Oil rigs
- Machine learning
- Finance

Minimization and Maximization, convention this course is for minimization
 Maximizing $f(x)$ will be seen as minimizing $-f(x)$.

$$\min_{x \in \mathbb{R}} f(x) \quad (1)$$

What characterizes an optimum?

-

$$\min_{x \in \mathbb{R}^n} f(x)$$

-

$$f \in \mathbb{C}^k$$

-

$$\frac{d}{dx} f(x^*) = 0$$

-

$$\frac{d^2}{dx^2} f(x^*) > 0$$

1.5 (Un)Constrained Optimization

Constraints are mathematical inequalities for the variables x_n for the desired task.

Unconstrained:

-

$$\min_{x \in \mathbb{R}^n} f(x)$$

Constrained:

-

$$\min_{x \in \mathbb{R}^n} f(x) \text{ s.t. } C_i(x) = 0, i \in \epsilon, C_i(x) \geq 0, i \in \mathbb{I}$$

.

-

$$C_i(x)$$

constrained function

- ϵ : Equality constraint
- $\mathbb{I}$: Inequality constraint

Feasible set:

-

$$\Omega = \{x \in \mathbb{R}^n \text{ s.t. } C_i(x) =$$

1.6 Linear programming

$$\min_{x \in \mathbb{R}^n \text{ s.t. }} \begin{cases} a_i^T x = b_i, i \in \epsilon \\ a_i^T x \geq b_i, i \in I \end{cases} \quad (2)$$

1.6.1 Farmer example, find max profit or min neg profit

1.7 Quadratic programming

$$\min x^T G x + c^T x \text{ s.t. } \begin{cases} a_i^T x = b_i, i \in \epsilon \\ a_i^T x \geq b_i, i \in I \end{cases} \quad (3)$$

1.8 Convexity

Convex set: A set is convex if

$$x, y \in S \implies \alpha x + (1 - \alpha)y \in S, \forall \alpha \in [0, 1] \quad (4)$$

Convex function: $f: S \rightarrow \mathbb{R}$ is a convex function if S is convex and *every* $x, y \in S : f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y), \forall \alpha \in [0, 1]$ "A line from point x to y should be below the function from $f(x)$ to $f(y)$ "

An optimization problem is convex if

- $f(x)$ is a convex function
- The feasible set is a convex set

1.8.1 Local and global solutions

x^* is a global solution if $f(x^*) \leq f(x) \forall x \in \Omega$. Let N be a neighborhood of x : x^* is a local solution if $f(x^*) \leq f(x) \forall x \in N \subseteq \Omega$

Ω is the set of all x values which fulfill the constraints.